

Unsupervised Machine Learning for Student Behaviour Analysis

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Introduction:

This project uses the Student Performance – Portuguese dataset from the UCI Machine Learning Repository. The goal is to group (cluster) students into different profiles based on their behaviour and background, not to predict a grade.

We remove the grades and treat this as an unsupervised clustering problem. We compare three clustering methods: KMeans, Agglomerative Clustering, and DBSCAN.

This project follows the principles of unsupervised learning, where the goal is not to predict an output but to discover natural patterns or groupings within the data without using labelled outcomes.

Dataset and project goal:

The dataset contains students from two Portuguese secondary schools and includes demographic, social and school-related features. Examples of features are: school, sex, age, address (urban/rural), family size, parents' education, study time, number of past class failures, absences, and alcohol consumption on workdays (Dalc) and weekends (Walc).

In this project, the goal is to find groups of students with similar profiles using variables such as studytime, failures, absences, Dalc and Walc. These groups can help teachers understand which students are more at risk and which are more stable.

The selected variables (studytime, failures, absences, Dalc and Walc) focus specifically on behavioural and engagement-related aspects, which aligns with the objective of clustering: grouping students based on similar characteristics rather than predicting academic outcomes.

Exploration (EDA):

First, the dataset is loaded and basic information is checked: number of rows and columns, feature names, and number of missing values. The total number of missing values is 0, so we do not need to impute or drop rows for this dataset.

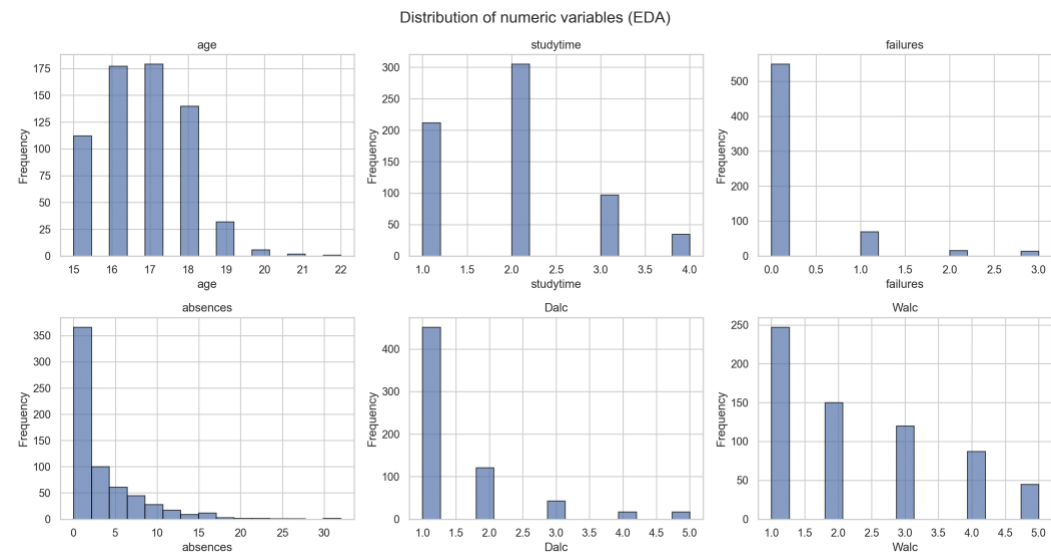


Figure 1. Histograms of age, studytime, failures, absences, Dalc and Walc

You created histograms for age, studytime, failures, absences, Dalc, and Walc. These histograms show how the values are distributed. For example, most students have no past failures, but a small number have many failures; alcohol consumption during the week is usually low, but higher on weekends.



Figure 2. Correlation heatmap between age, studytime, failures, absences, Dalc and Walc

You also created a correlation heatmap for these numeric variables. This figure shows how strongly variables move together. For example, Dalc and Walc are usually positively correlated, because students who drink during the week also often drink on weekends. These EDA steps match the assignment requirements: visualising distributions, looking for relationships, and checking for missing values.

After one-hot encoding, the dataset becomes high-dimensional, which is important because high dimensionality can make clustering more difficult due to challenges in measuring similarity.

Data Preprocessing:

First, the three grade columns G1, G2, and G3 are dropped, because the focus is on behaviour clusters, not final grades.

Next, all categorical variables (such as school, sex, address, parents' jobs, etc.) are converted to numeric form using one-hot encoding (`pd.get_dummies` with `drop_first=True`). This creates binary columns for each category and avoids problems with algorithms that use distances.

Finally, all features (after encoding) are standardised with `StandardScaler`. Each feature is scaled to have mean 0 and variance 1. This is important so that variables with large scales (like absences) do not dominate variables on small scales (like Dalc and Walc).

Standardising the features is essential because distance-based clustering algorithms, especially KMeans and Agglomerative Clustering, rely heavily on Euclidean distances and are therefore sensitive to differences in feature scales.

Clustering methods:

KMeans clustering:

KMeans is run with different values of *k* (number of clusters) from 2 to 10. For each *k*, the algorithm is fitted on the scaled data and labels are predicted. The silhouette score is calculated to measure how well separated and compact the clusters are. In addition to the silhouette score, KMeans can also be evaluated using inertia (Within-Cluster Sum of Squares), which measures cluster compactness, although in this project we prioritised silhouette-based evaluation. The best *k* is chosen as the one with the highest silhouette

score. Then a final KMeans model is trained with this best k , and the resulting labels are stored in a new column `cluster_kmeans` in the DataFrame.

Agglomerative clustering:

Agglomerative Clustering is another method that builds clusters by merging points step by step. Agglomerative Clustering was performed using Ward linkage, which minimises within-cluster variance and generally produces compact, well-separated clusters. As with KMeans, the number of clusters k is tested from 2 to 10, and for each k a silhouette score is computed. The best number of clusters for Agglomerative Clustering is also the one with the highest silhouette score. A final model is then fitted with this value of k , and the labels are added as `cluster_agglo`.

DBSCAN clustering:

DBSCAN is a density-based method that does not require choosing k directly. Instead, it needs two hyperparameters: the neighbourhood radius `eps` and the minimum number of points `min_samples`. A small grid of values is tested: `eps` in {0.5, 0.7, 1.0, 1.5} and `min_samples` in {3, 5, 10}. For each pair, DBSCAN is fitted and the number of clusters (excluding noise points with label -1) is checked. Only cases with at least 2 clusters are kept, then a silhouette score is computed and the best combination of `eps` and `min_samples` is selected. The final DBSCAN model uses these best parameters, and its labels are stored in `cluster_dbscan`. Noise points are labelled -1.

DBSCAN typically performs best in low-dimensional datasets with clear dense regions; in high-dimensional data, density becomes harder to define, which explains its difficulty in forming stable clusters in this project

Cluster Visualisation:

A line plot shows the silhouette score for each k for both KMeans and Agglomerative Clustering. The best k values are marked with vertical dashed lines. This graph helps to justify the choice of the number of clusters for both algorithms.

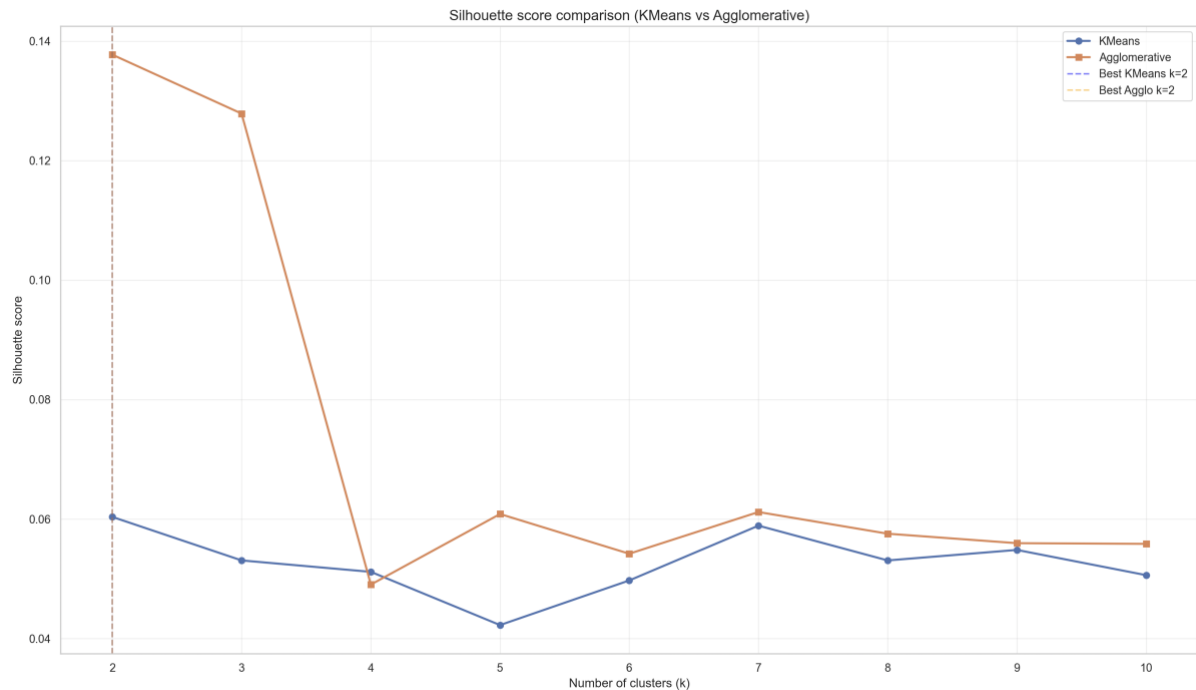


Figure 3. Silhouette score versus number of clusters k for KMeans and Agglomerative Clustering.

A dendrogram is created on a sample of about 100 students using a hierarchical clustering linkage matrix. It shows how points are merged into clusters and at what distance this happens. This figure helps to see visually how many big groups exist in the data.

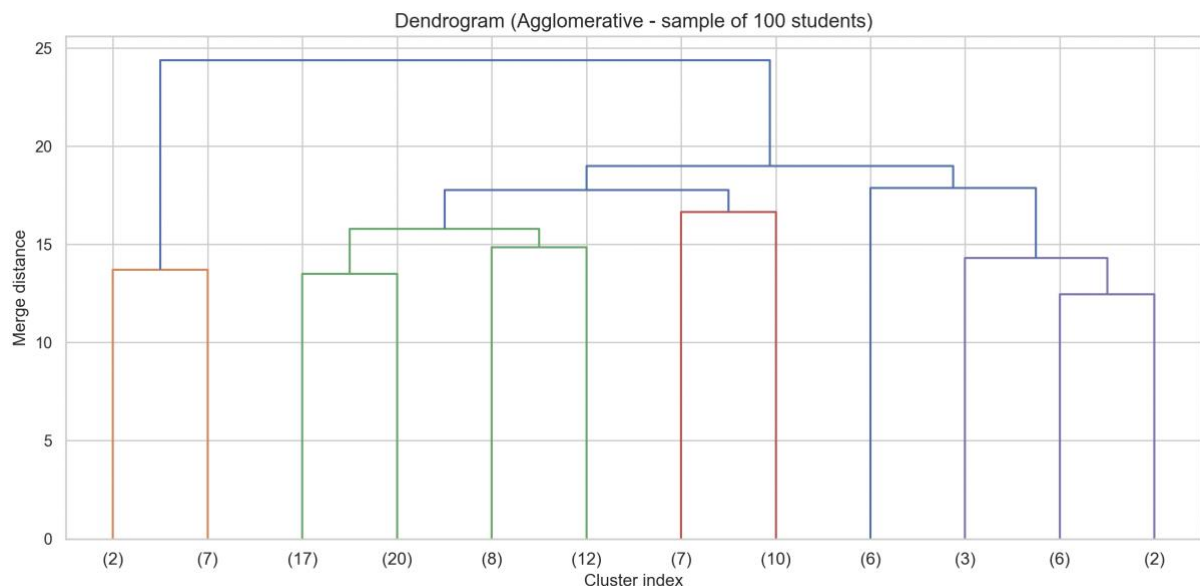


Figure 4. Dendrogram for hierarchical (Agglomerative) clustering on a sample of students.

Boxplots are drawn for key variables (studytime, failures, absences) for each cluster of KMeans (Figure 5) and for each cluster of Agglomerative Clustering (Figure 6). These plots

show differences between groups: for example, one cluster may have higher study time and fewer failures, while another may show more absences and more failures.

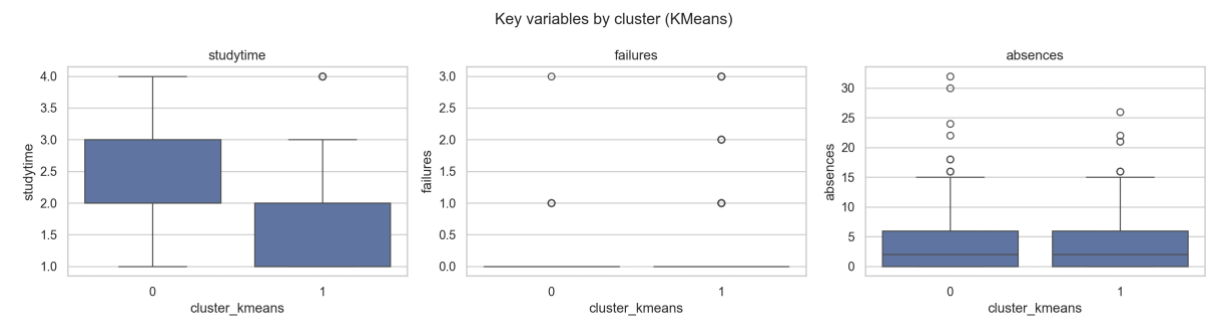


Figure 5. Boxplots of studytime, failures, and absences by KMeans cluster

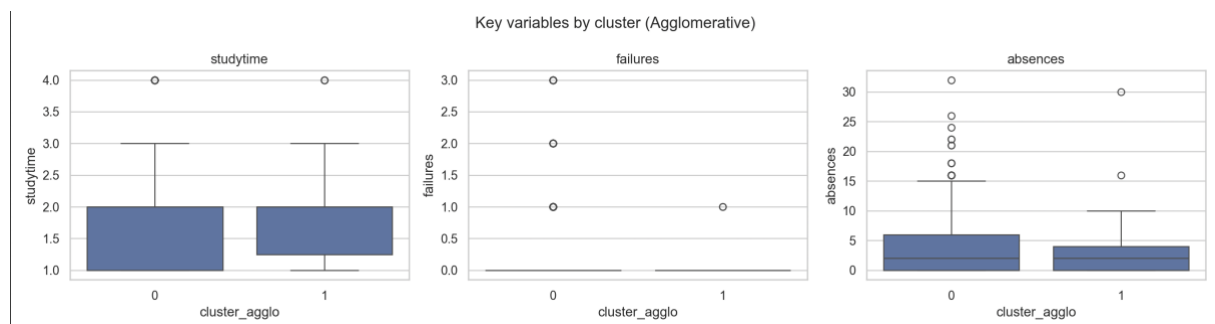


Figure 6. Boxplots of studytime, failures, and absences by Agglomerative cluster

A Principal Component Analysis (PCA) reduces all features to two components (PC1 and PC2). Two 2D scatter plots are created: one coloured by KMeans cluster labels, and one coloured by Agglomerative cluster labels. These plots let us see how separated the clusters are in a 2D projection.

PCA was applied purely for visualisation purposes, as reducing high-dimensional data to two principal components allows cluster separation to be inspected in 2D without influencing the clustering process itself.

If DBSCAN is used, a PCA scatter plot coloured by DBSCAN labels can also be plotted, where points with label -1 are noise. This makes it easy to see how many points DBSCAN treats as noise and whether the dense clusters are clearly visible.

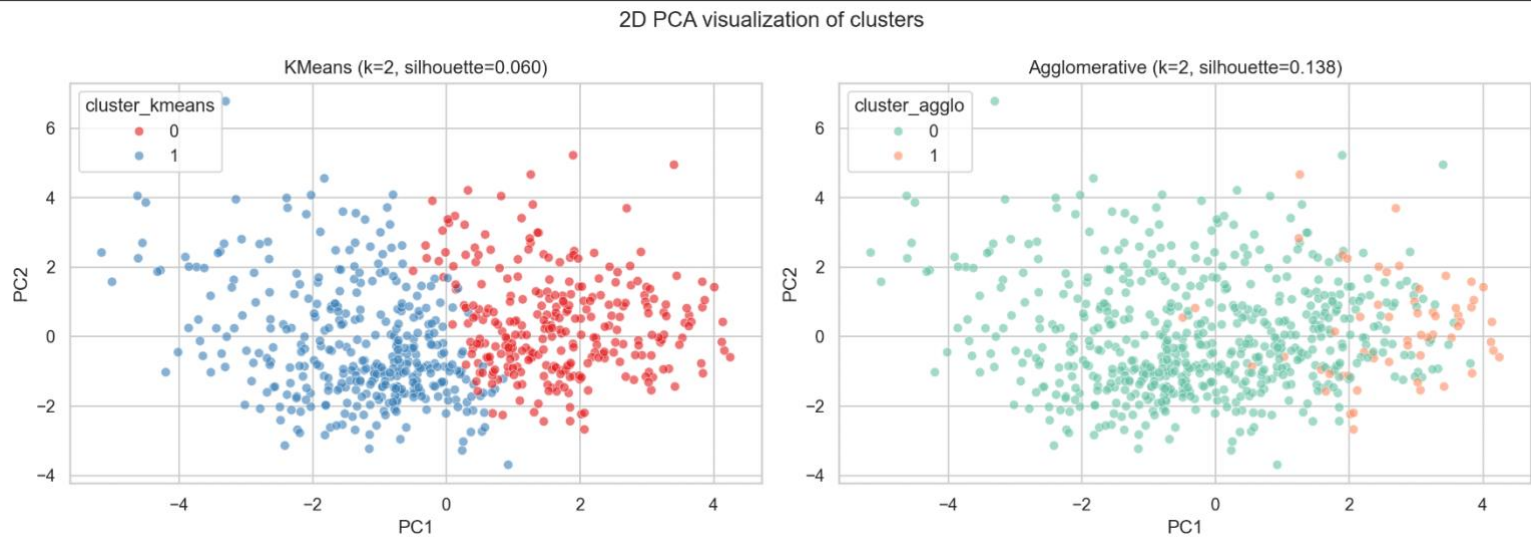


Figure 7. PCA 2D scatter plot coloured by KMeans clusters.

Results and interpretation of clusters:

For KMeans with $k=2$, cluster 0 contains 283 students and cluster 1 contains 366 students. Cluster 0 has higher average studytime (about 2.08 vs 1.82 on the 1–4 scale), almost no previous failures (≈ 0.05 vs 0.35), and slightly higher absences, while alcohol use is similar in both clusters, so cluster 0 can be interpreted as a somewhat more studious and lower-risk group. Cluster 1 therefore corresponds to a larger, more mixed group with lower studytime and more failures, indicating higher academic risk.

With Agglomerative clustering and $k=2$, cluster 1 is a small high-achievement group of about 58 students with higher studytime (≈ 2.05 vs 1.92), almost no failures (≈ 0.03 vs 0.24), slightly fewer absences, and lower alcohol consumption than the majority cluster 0. The majority cluster 0 gathers 591 students with more heterogeneous and often weaker academic profiles, which matches the idea of one compact “strong” group and one broad at-risk group.

Comparing silhouette scores, KMeans reaches only about 0.06, while Agglomerative achieves about 0.14, so the hierarchical model provides better-separated clusters on this dataset even though the overall clustering quality remains modest. For DBSCAN, no tested combination of `eps` and `min_samples` produced stable multi-cluster solutions with a good silhouette score, and many points were labelled as noise, which suggests that DBSCAN is less suitable for this high-dimensional dataset

Insights and recommendations:

The clustering suggests that some students share a profile of good study habits, few absences, and low alcohol use, while others share a profile of higher absences, more failures, and higher alcohol use. These patterns match previous analyses of student performance datasets and show that lifestyle and engagement variables are strongly linked to academic risk.

Schools could use these insights to design targeted actions: extra help and mentoring for high-risk clusters, awareness sessions about alcohol and absenteeism, or positive reinforcement for students in low-risk clusters.

Limitations and future work:

There are several limitations. In particular, even the best silhouette score (around 0.14) is low, which means that student profiles overlap strongly and the clusters should be interpreted as soft tendencies rather than clearly separated types. First, grades G1–G3 were removed from the data used for clustering, so we cannot directly measure how each cluster performs academically. Second, the data comes from self-reported questionnaires, which may contain bias or inaccurate answers, especially for sensitive topics like alcohol consumption. Future work could include using the clusters as features in a classification model that predicts the final grade or pass/fail status, to see how cluster membership relates to performance. It would also be useful to test other clustering methods, such as Gaussian Mixture Models or non-linear dimensionality reduction plus clustering (for example, t-SNE or UMAP followed by KMeans), to see if they can discover more detailed subgroups. These clusters should not be interpreted as causal categories but rather as data-driven groupings that reveal behavioural patterns without implying prediction or causality.